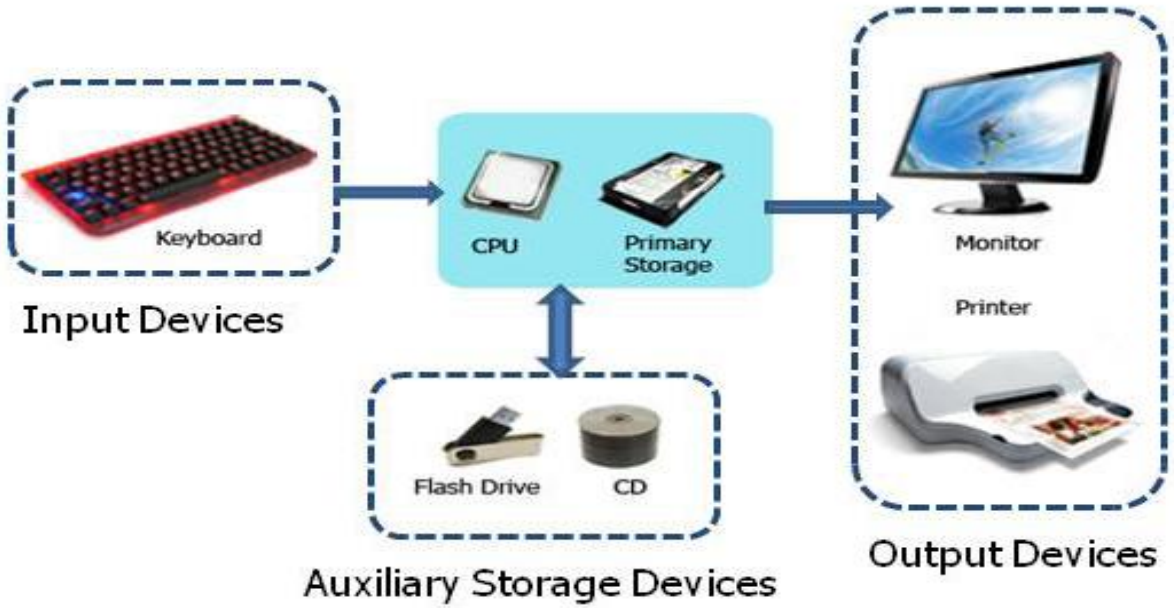
Digital Circuit and Logic Design



Chapter 12
Microcomputer and Microprocessor

Computer

Programmable machine designed to sequentially and automatically carry out a sequence of arithmetic or logical operations. The particular sequence of operations can be changed readily, allowing the computer to solve more than one kind of problem.



Computer Type



Mainframe (www.top500.org)
Sunway TaihuLight 93.86 PFLOPS, RaiseOS
10M cores Sunway SW26010, 1PB, 15.3MW
Mini computer
Micro computer (PC / Mac)
- Tower,Desktop,Notebook,Ultrabook

Tablet (iPad, Galaxy Tab)
PDA (Personal Digital Assistant) -> Smartphone
Embedded System (Raspberry PI, Arduino)
Microcontroller (ARM, AVR, ESP32)
D-Wave 2x Quantum Computer 15.9kW

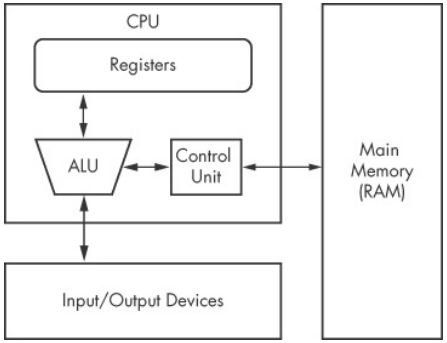


Microcomputer

- Apple II (1977, MOS6502-8 bit home computer)



- IBM PC (1981, x86 architecture)





- Apple Macintosh (Motorola 68000 -> Intel x64)

PC - Personal Computer





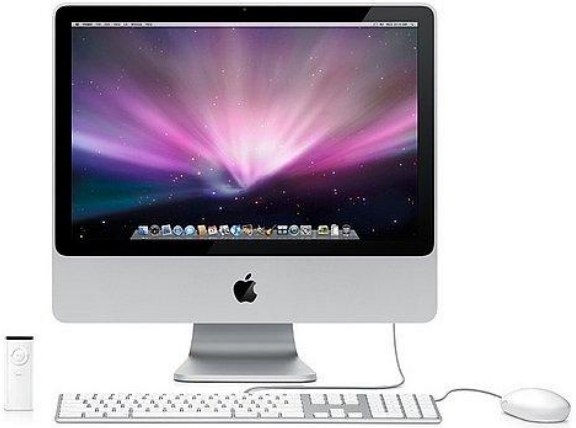




Apple Macintosh



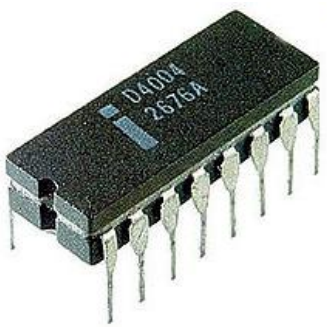








Evolution of Intel® Microprocessor





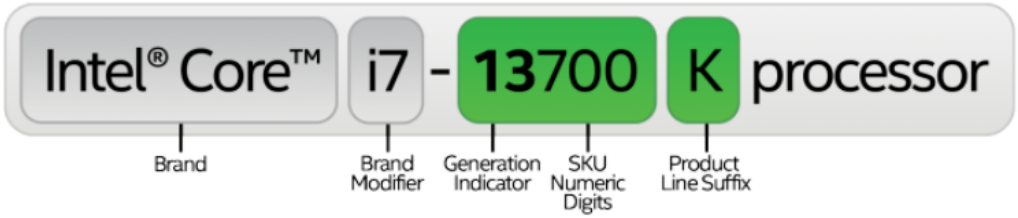
1971: Intel uP 4004

Used for calculator
2,300 transistors
400 –740 kHz
4-bit word size
16-pin DIP package

2017: Intel Core i9-7980XE (Extreme Edition)

1.4 billions transistors 165W
x64 18 cores 2.6 GHz
24MB cache
LGA socket

Intel® Microprocessor



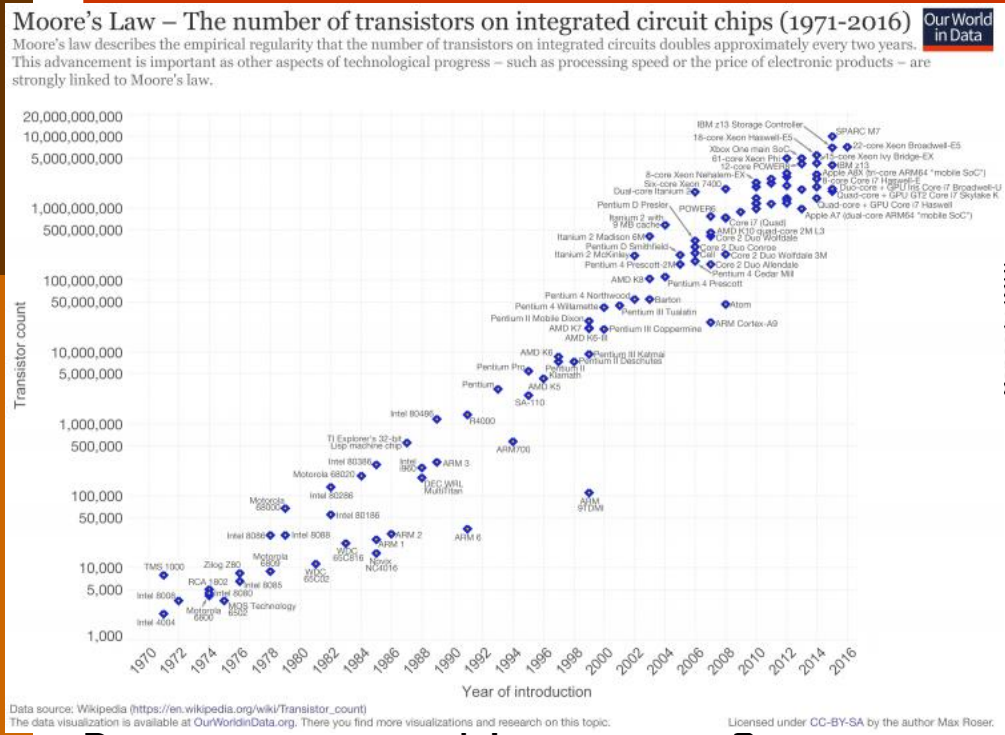
Brand Modufier: the Intel® Core™ processor series includes the brand modifiers i3, i5, i7, and i9. Higher brand modifier numbers offer a higher level of performance and, in some cases, additional features

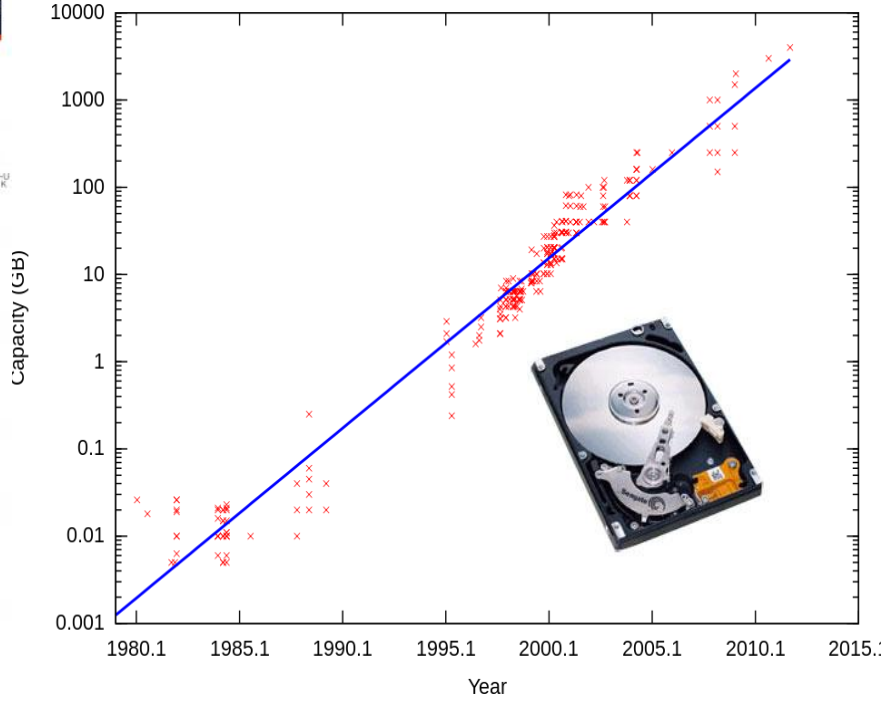
13th Generation Intel® Core™ Processor Family advances performance hybrid architecture with up to eight P-cores and up to 16 E-cores offering a superior creating and gaming experience. These processors deliver highly flexible architecture and industry-leading tools for the ultimate performance customization.



Form/Function Type/Segment	Suffix	Optimized/Designed For
Desktop	K	High performance, unlocked
	F	Requires discrete graphics
	S	Special edition
	T	Power-optimized lifestyle
	X/XE	Highest performance, unlocked

Moore's Law





Processor speed increases 2x every 18 months

Disk size increases 2x every year



x86/x64 CPUs

x86	8086/8088 80186/80286	NEC V20
IA-32	80386/80486 Pentium Celeron	AMD386 / AMD486 AMD K5/K6 Athlon
IA-64 or x64	Xeon Core i3, i5, i7 8th Gen (986), i9 Pentium Celeron Atom	Opteron Ryzen AMD-FX, Athlon X4 Sempron AMD APU

Unit of Measurement in Computer System

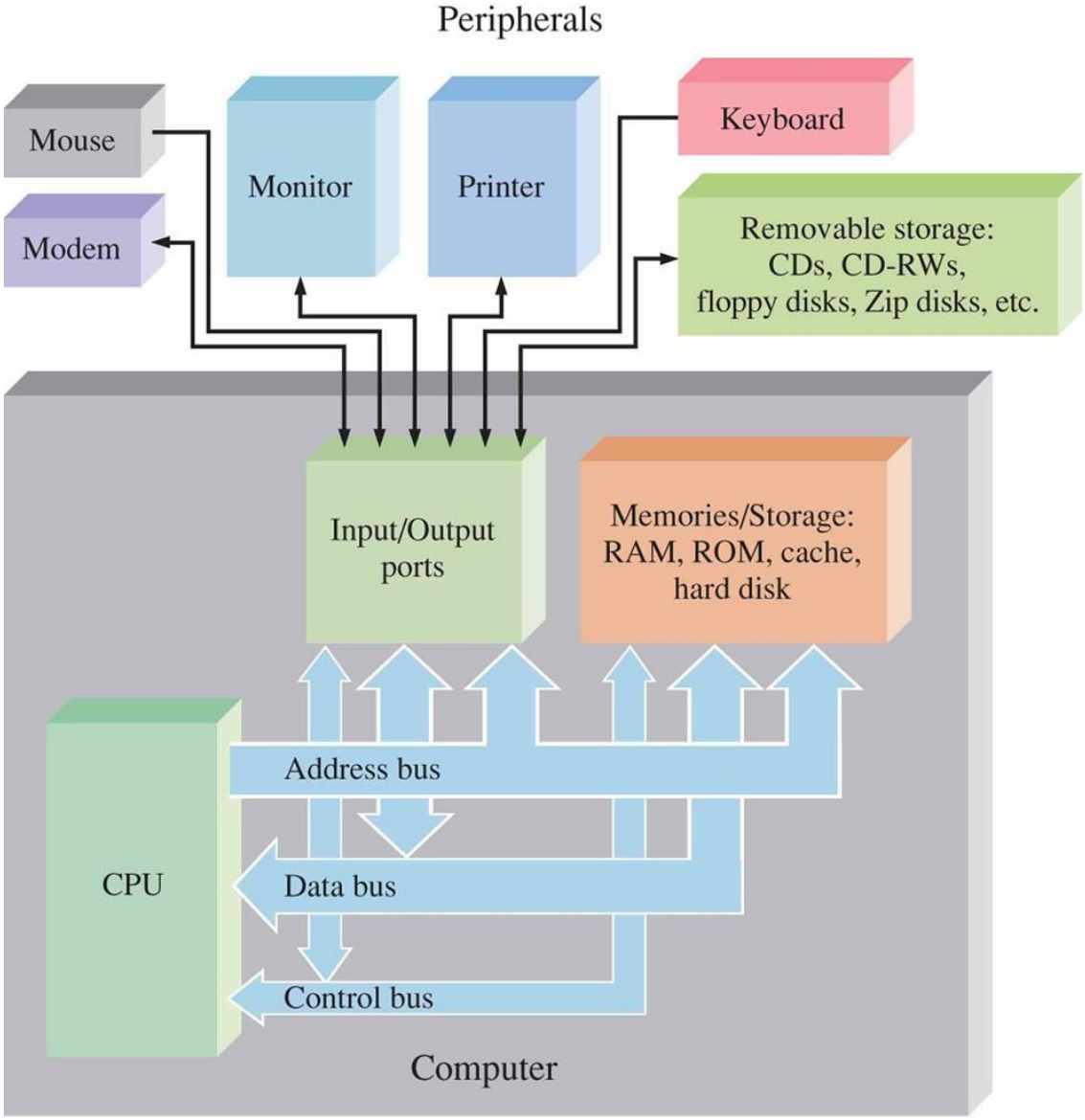
Name	Abbr	Factor	SI size
Kilo	k	2^{10} = 1,024	10^{3} = 1,000
Mega	M	2^{20} = 1,048,576	10^{6} = 1,000,000
Giga	G	2^{30} = 1,073,741,824	10^{9} = 1,000,000,000
Tera	T	2^{40} = 1,099,511,627,776	10^{12} = 1,000,000,000,000
Peta	P	2^{50} = 1,125,899,906,842,624	10^{15} = 1,000,000,000,000,000
Exa	E	2^{60} = 1,152,921,504,606,846,976	10^{18} = 1,000,000,000,000,000,000
Zetta	Z	2^{70} = 1,180,591,620,717,411,303,424	10^{21} = 1,000,000,000,000,000,000,000
Yotta	Y	2^{80} = 1,208,925,819,614,629,174,706,176	10^{24} = 1,000,000,000,000,000,000,000,000

Ex. A 1 TB harddisk
Manufacturer uses TB in SI unit, so the harddisk has 1,000,000,000,000 B capacity
MS Windows uses TB as 2^{40}, so the Windows can see the harddisk around 909 GB

A Unix® based operating system like macOS X® or Linux® uses decimal bytes when reporting storage space, so a 480GB SSD will show up as 480GB in Mac Disk Utility for instance. The Windows® OS on the other hand uses binary bytes, so 1,024 bytes per Kilobyte, 1,024 KB per Megabyte, and so on. This means that when you install a 480,000,000,000 bytes storage drive into a Windows® computer, that computer converts the number of bytes into gigabytes by dividing by 1024 all the way up through the scale, not by dividing by 1,000. Doing the math, this is what we end up with:

480,000,000,000 Bytes / 1,024 = 468,750,000 actual Kilobytes
468,750,000 KB / 1,024 = 457,764 actual Megabytes
457,764 MB / 1,024 = 447 actual Gigabytes

Basic block diagram of a computer system



CPU - Central Processing Unit

- **ALU (Arithmetic Logic Unit)**

  The ALU performs computational functions such as Add, Subtract, AND, OR, Compare, Increment, and Decrement.

- **Control Logic**

  The control logic decodes and executes the program. It also controls the memory, input, and output operation of the microprocessor

- **Registers**

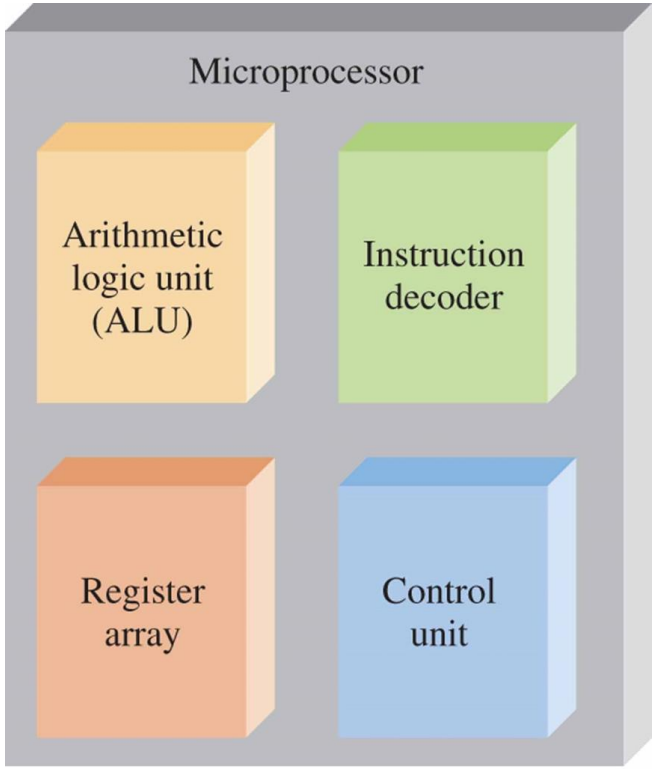
  The registers (fast memory, local to CPU) use to store temporary data and instructions

Microprocessor

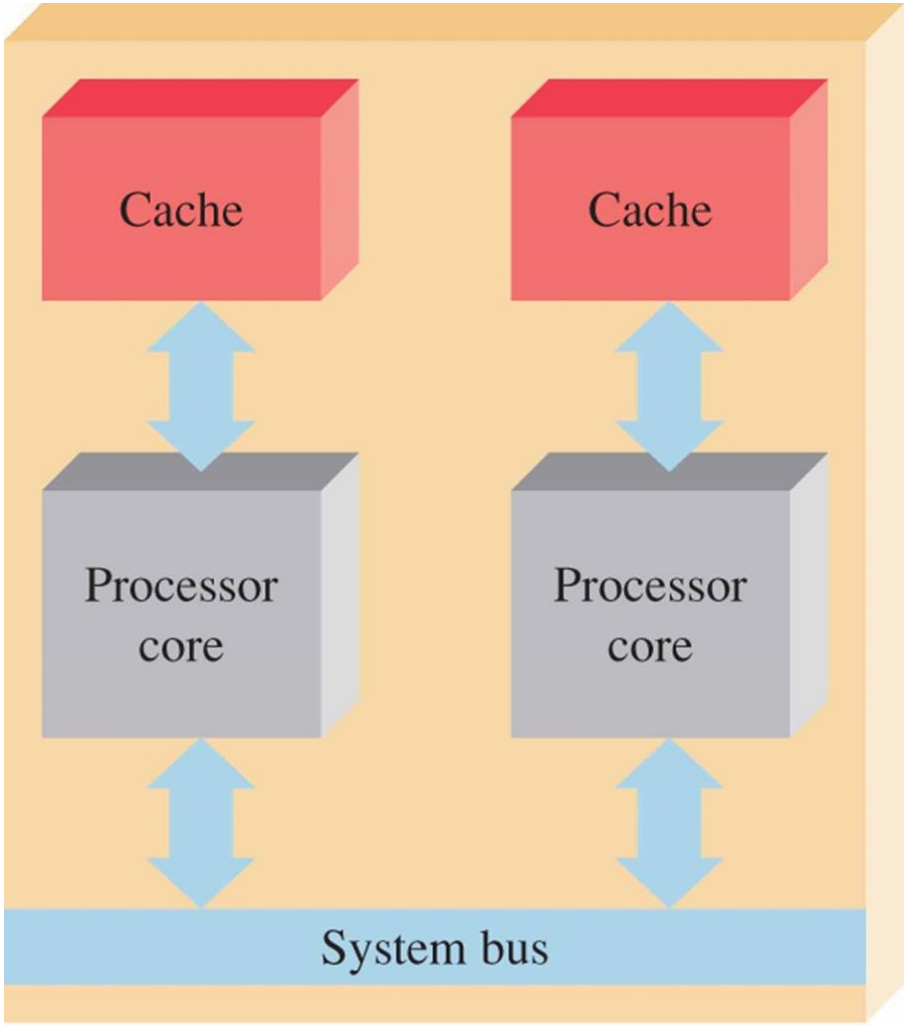
Very large-scale integrated circuit (VLSI) that uses the architecture of the general-purpose digital computer.

Microprocessors are based on the Von Neumann model of a stored program computer

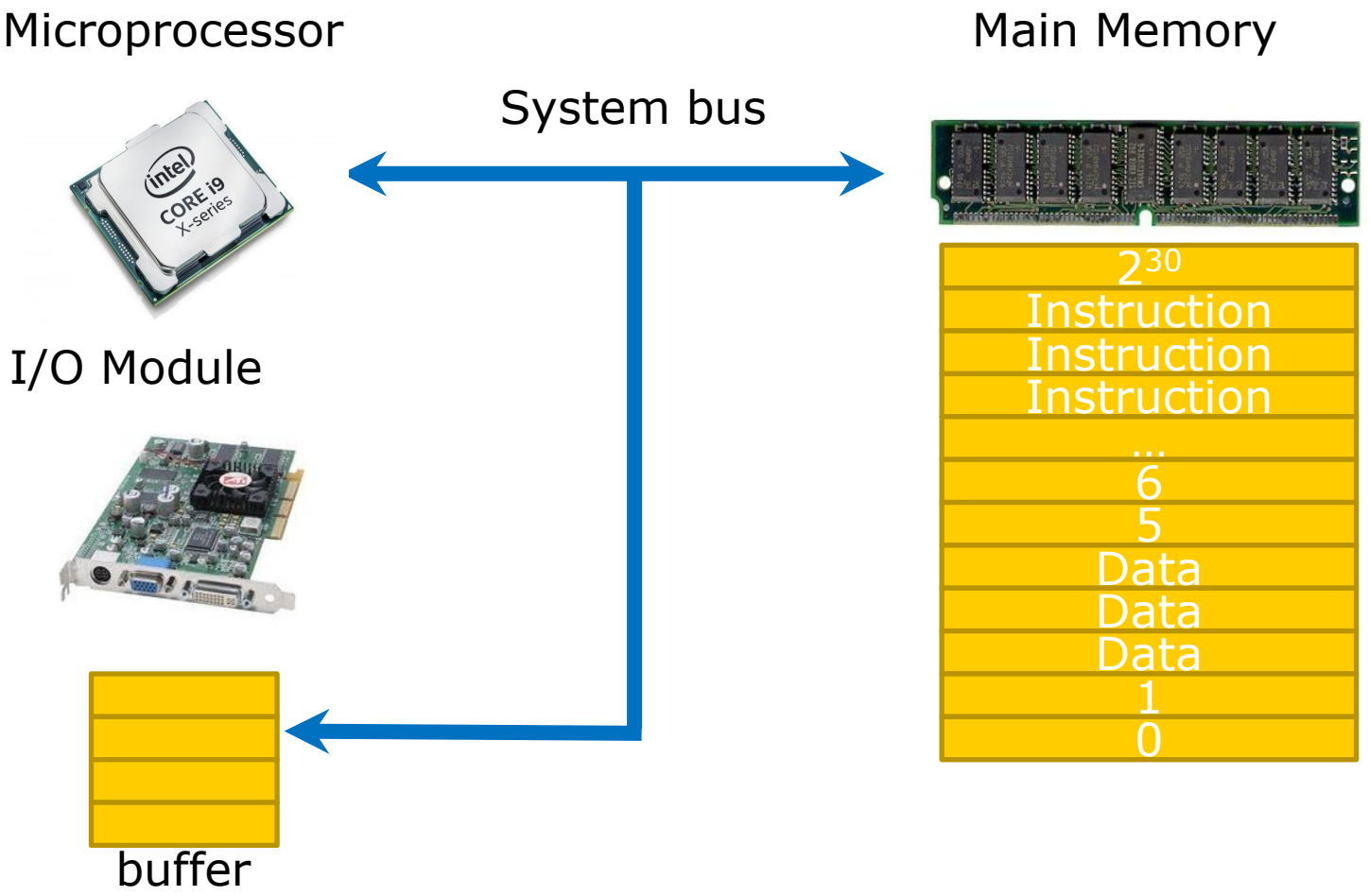
The stored program computer, a microprocessor's program is stored in memory along with its data



Dual-core processor



Main Components of a Micro Computer



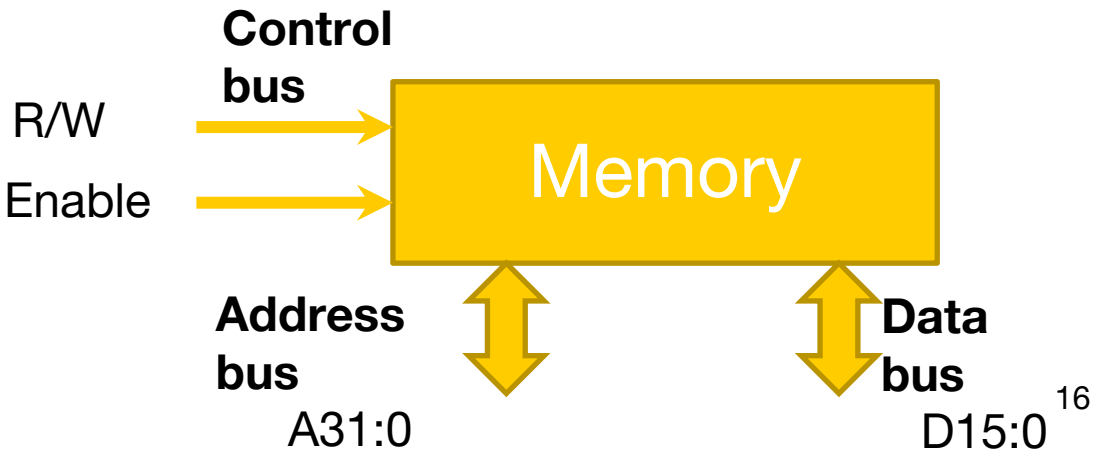
Memory

The communication between microprocessor and memory has 3 types of signals (buses).

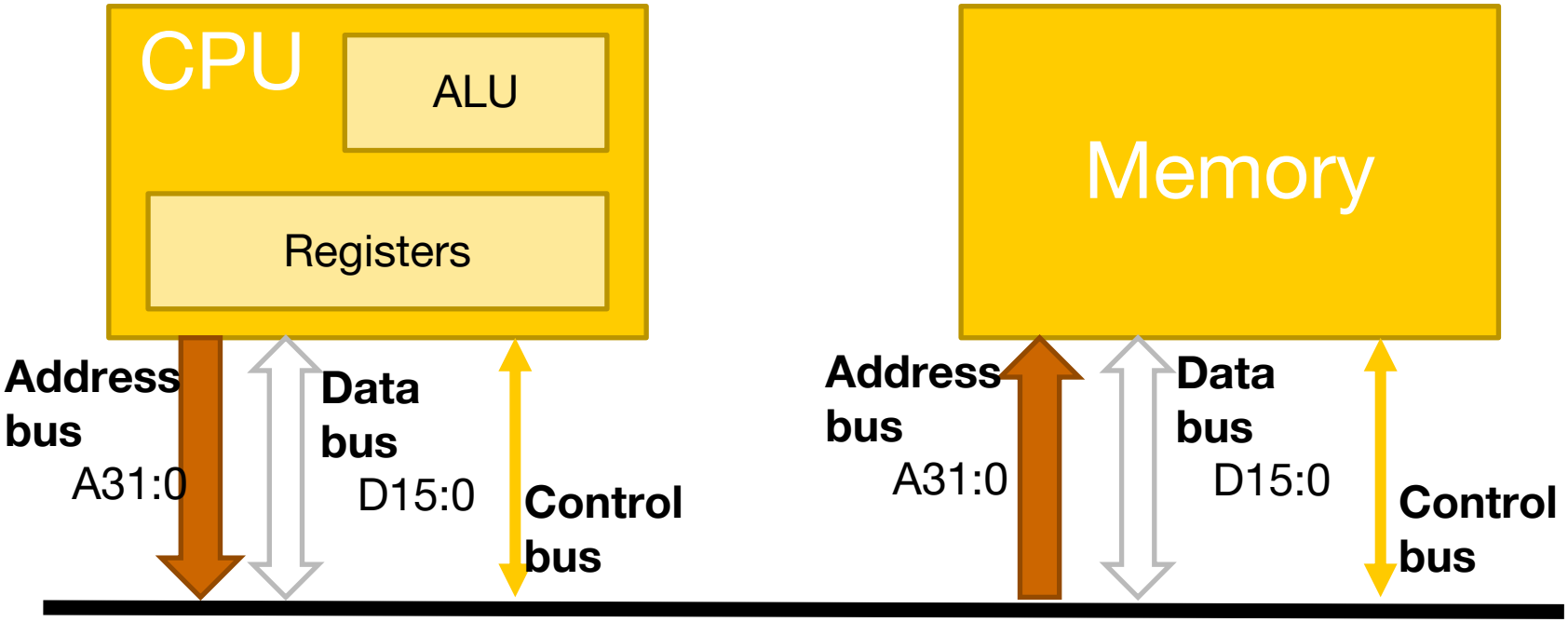
- Address bus – determines the location of memory
- Data bus – carries the contents of the location
- Control bus – governs the information transfer

The width of the address bus determines the size of memory. (how many location)

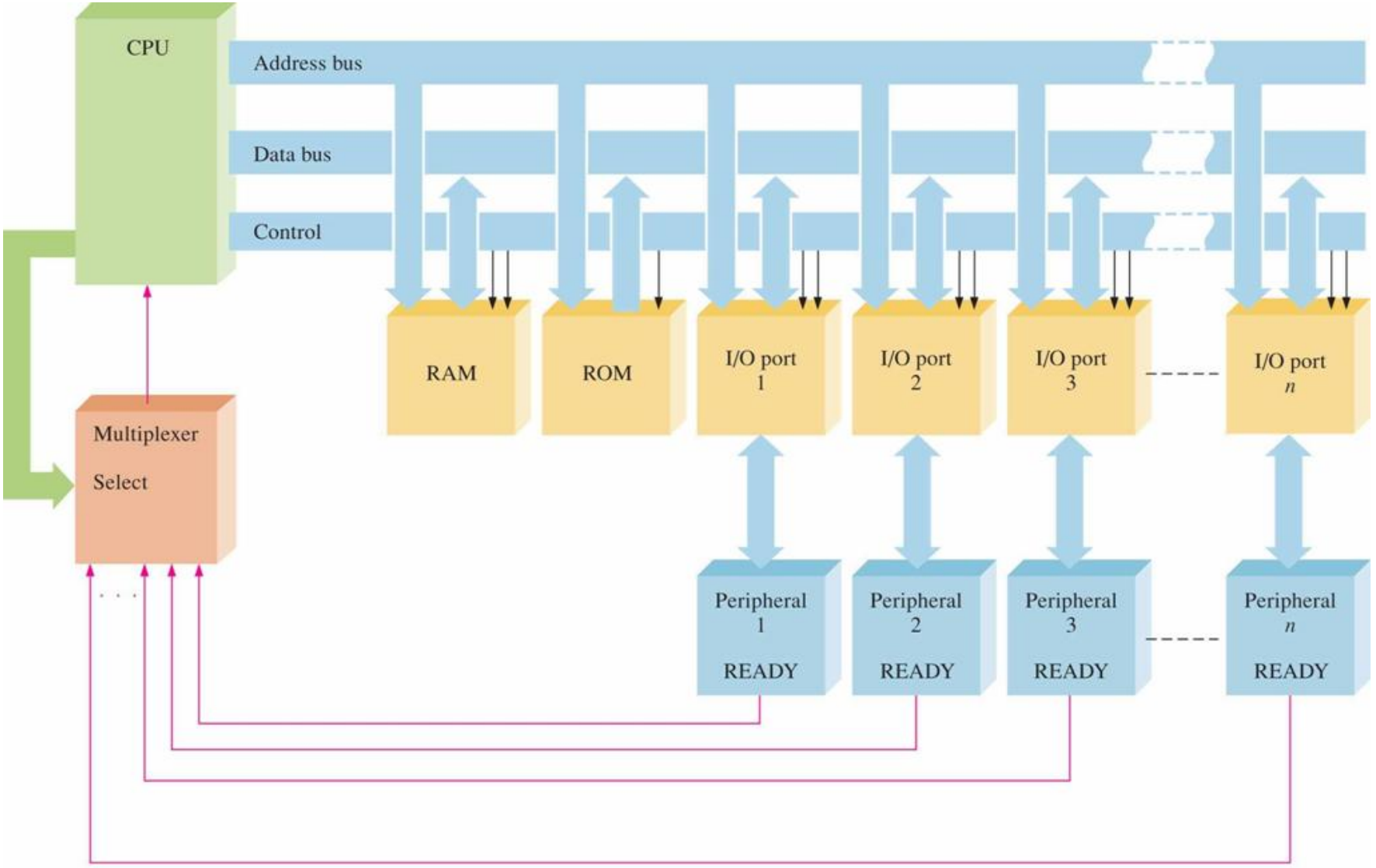
The width of the data bus determines the size of content. (how many bits can each location store)



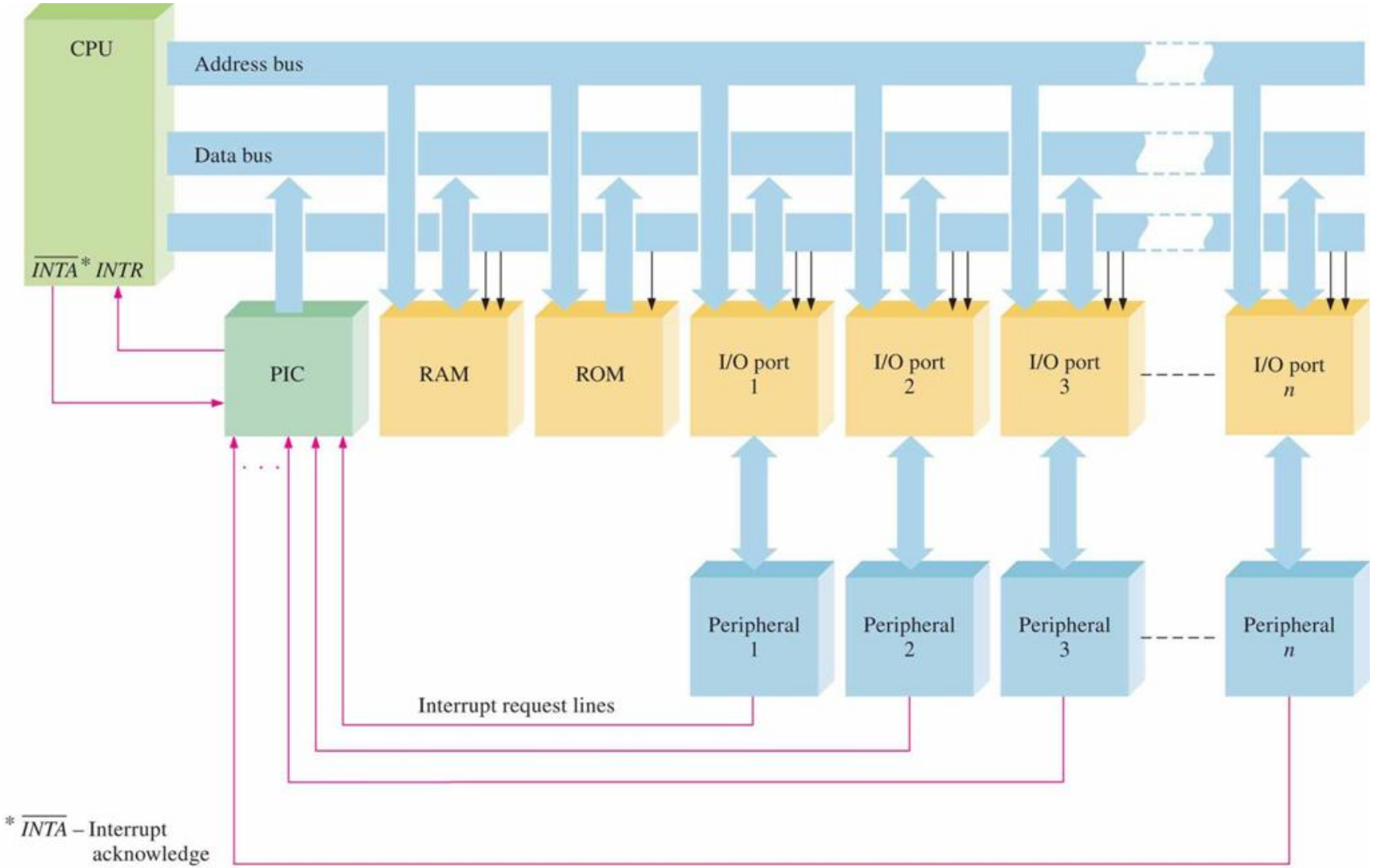
Memory, Registers, and ALU



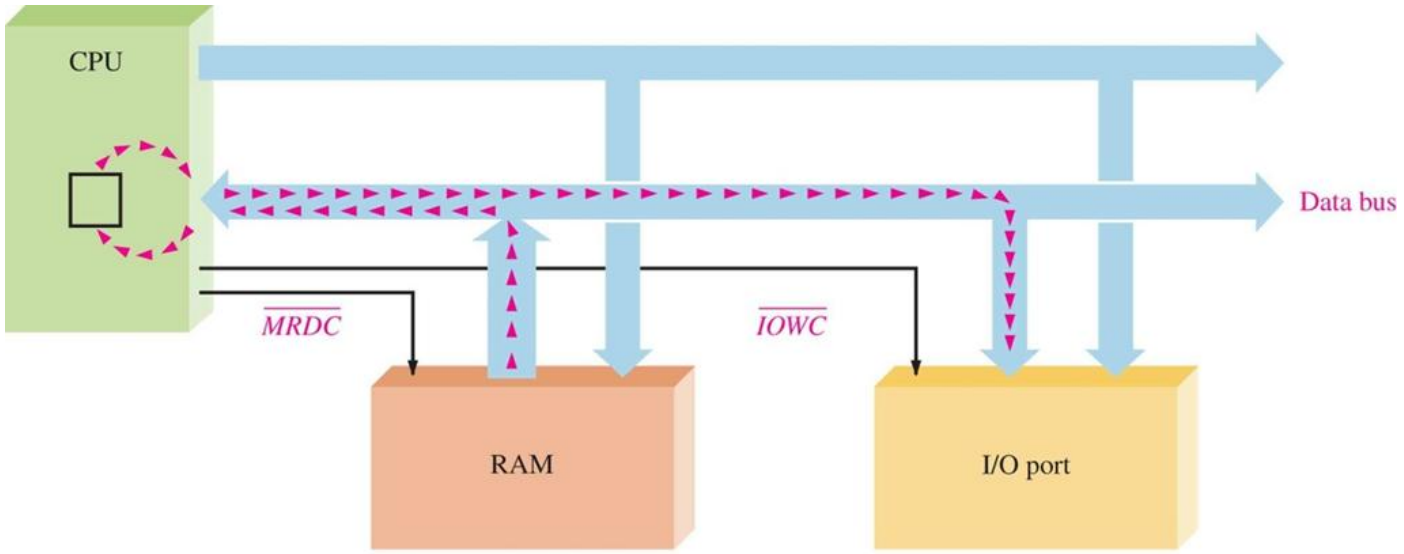
Polled I/O configuration

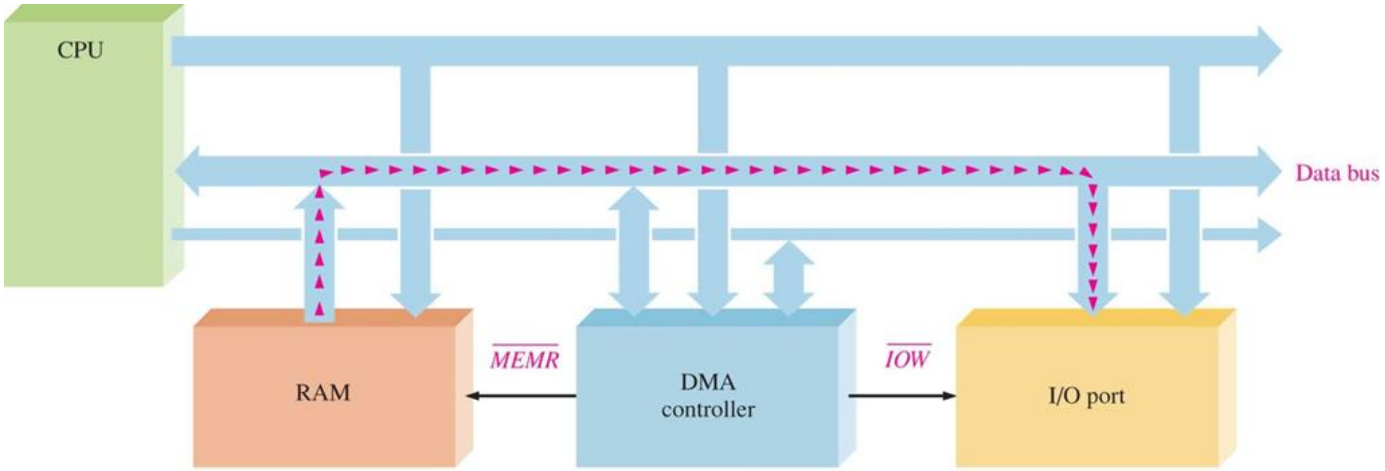


Interrupt-driven I/O configuration

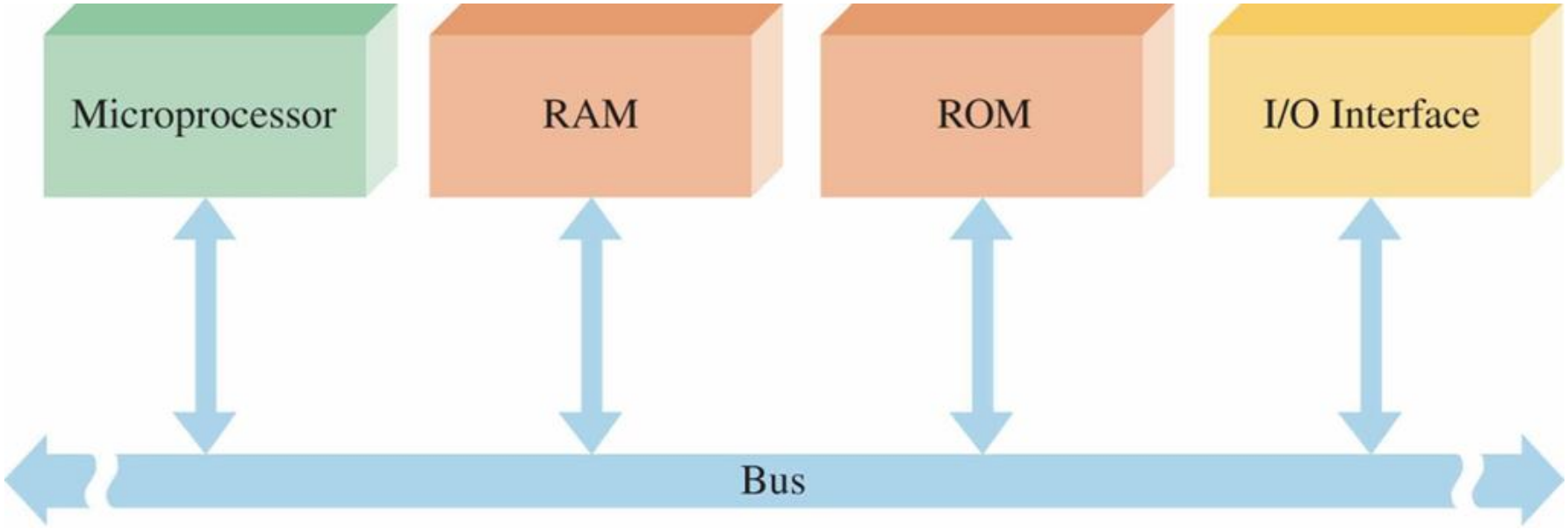


Memory I/O transfer handled by the CPU and DMA

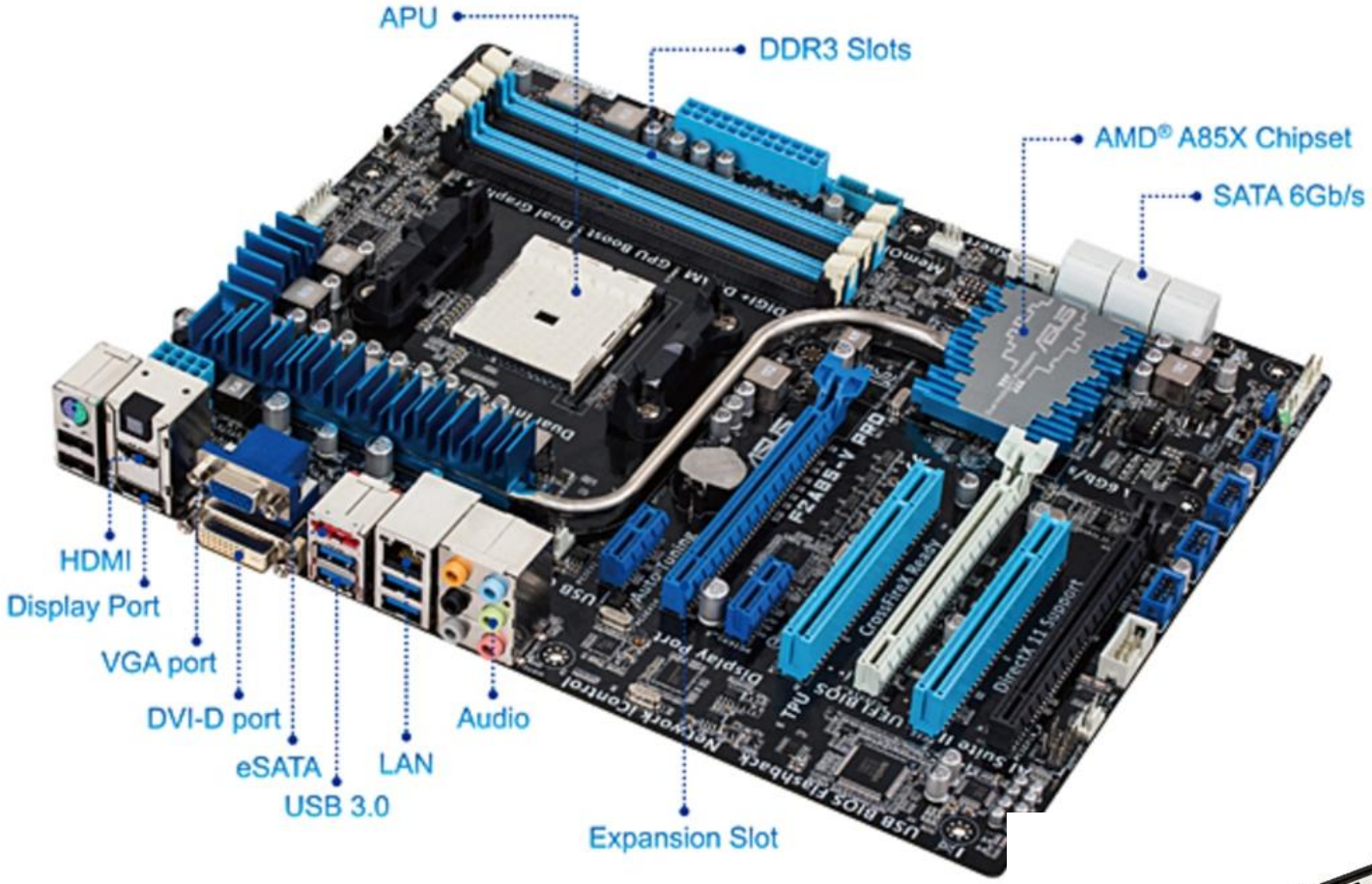




The interconnection of microprocessor-based system



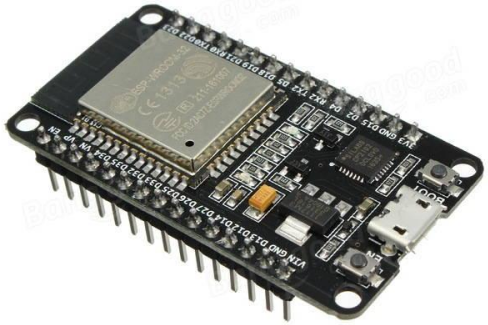
x86/x64 Main board



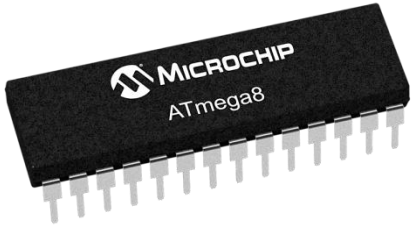


Embedded System



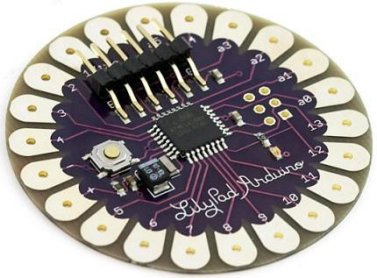


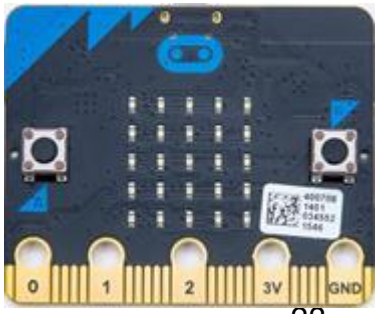












Memory

-Primary Memory or Main Memory

ROM – Read Only Memory (Non-volatile)
RAM – Random Access Memory (volatile)
NVRAM - Non-volatile Random-Access Memory

-Secondary Memory or Storage Memory

Magnetic Tape
Floppy Disk
Hard Disk
Optical Disk
Thumb Drive

RAM - Random Access Memory

SRAM – Static RAM



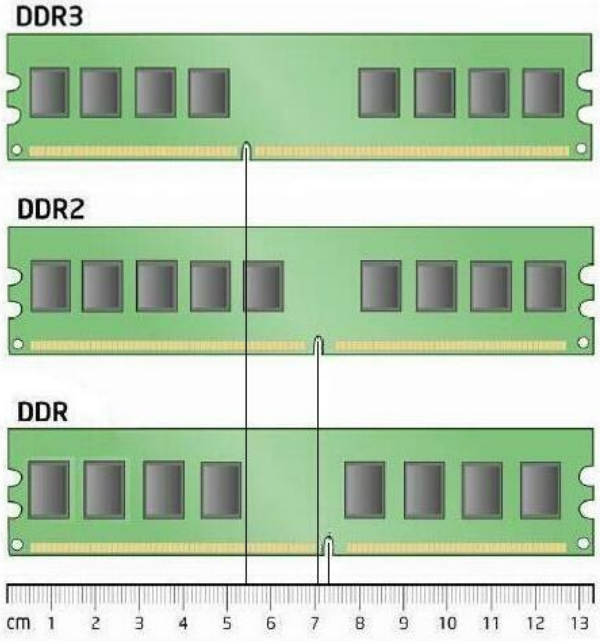
DRAM – Dynamic RAM

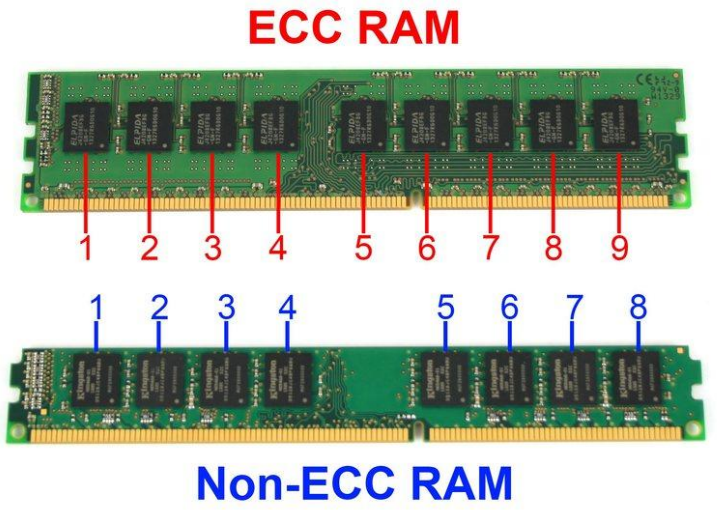
In development
T-RAM Thyristor RAM
Z-RAM Zero-capacitor RAM

Dynamic RAM

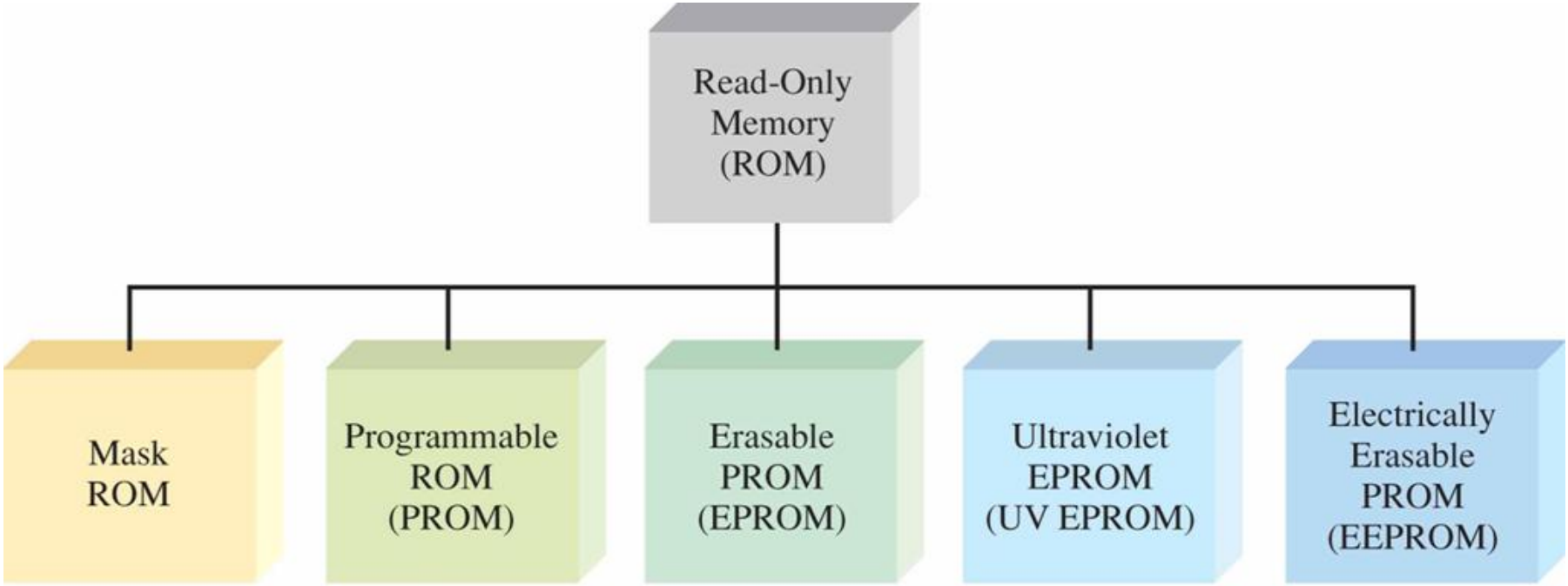
- SDRAM
- DDR SDRAM
- DDR2
- DDR3
- DDR4
- ECC (Error-Correcting Code)

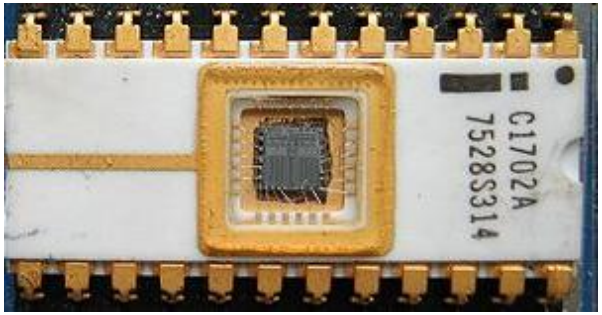






ROM


Read-Only Memory (ROM)
Mask ROM
Programmable ROM (PROM)
Erasable PROM (EPROM)
Ultraviolet EPROM (UV EPROM)
Electrically Erasable PROM (EEPROM)


C1702A
7528S314

NVRAM

- Flash memory (Toshiba)
- Solid-state storage

NAND Flash Memory

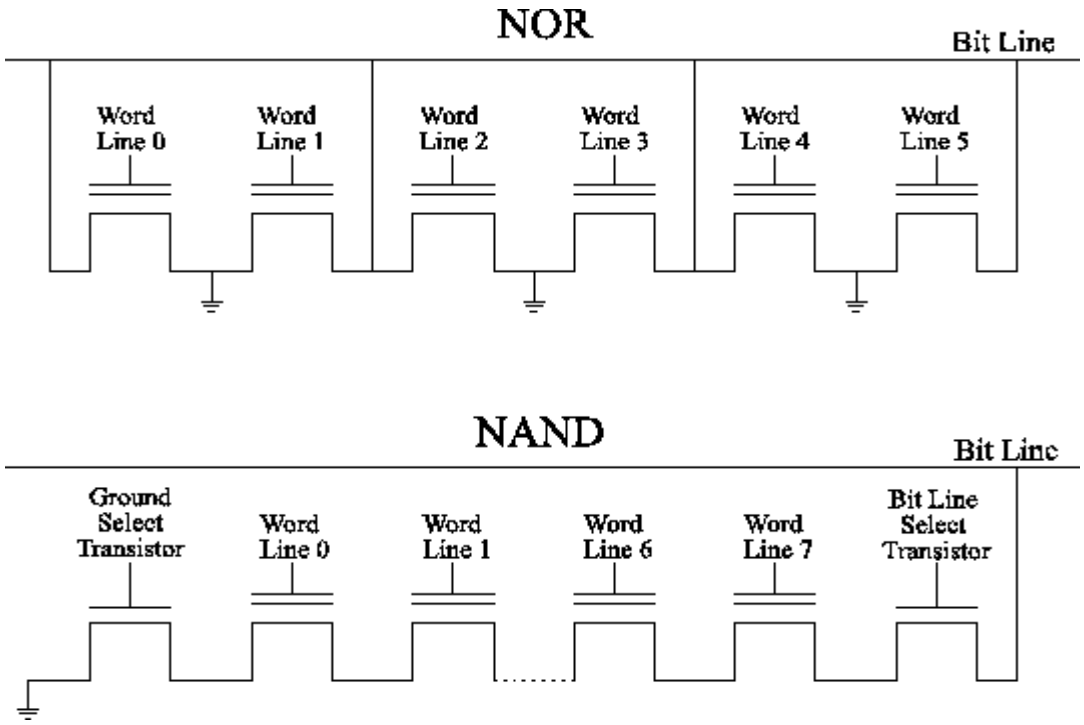
- Reads and writes in **sequential mode**, handling data in small, block sizes (“pages”).
- Commonly found in SSD, audio and video Flash media devices, cell phones (for data storage) and other devices where data is generally written or read sequentially.
- Less expensive than NOR Flash memory, and can accommodate more storage capacity in the same die size.





NOR Flash Memory

- Provides high-speed **random-access capabilities.**
- Allows the retrieval of data as small as a single byte. NOR Flash excels in applications where data is randomly retrieved or written. NOR Flash is most often found built into cellular phones (to store the phone's operating system)



Display Card





Nvidia
AMD(ATI)

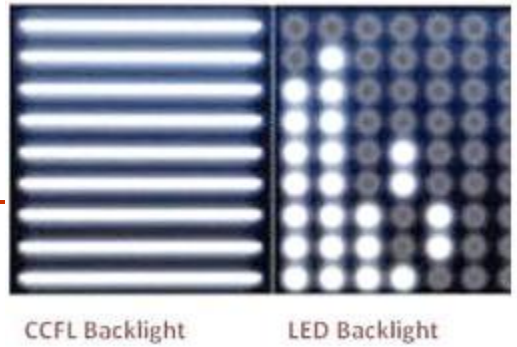
VGA / S-Video / DVI

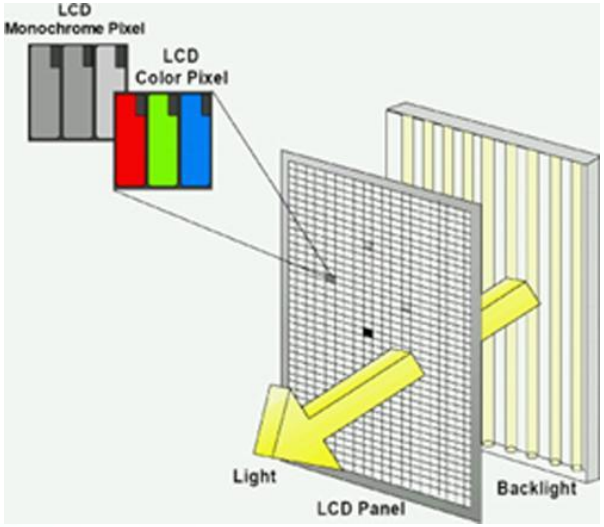
GPU - Graphics Processing Unit

APU - Accelerated Processing Unit
AMD Fusion

Output Unit

- CRT (Cathode Ray Tube)
- LCD
  - CCFL(Cold Cathode Fluorescent Lamp)
  - LED
- OLED
- e-Ink
- Printer
  - Dot-Matrix / Inkjet / Thermal / Laser
- Plotter
- Loud speaker (7.1 surround sound)









Input Unit

- Trackball
- Microphone
- Touch screen panel
- Digitizer
- Tablet







Network Interface Cards - NICs





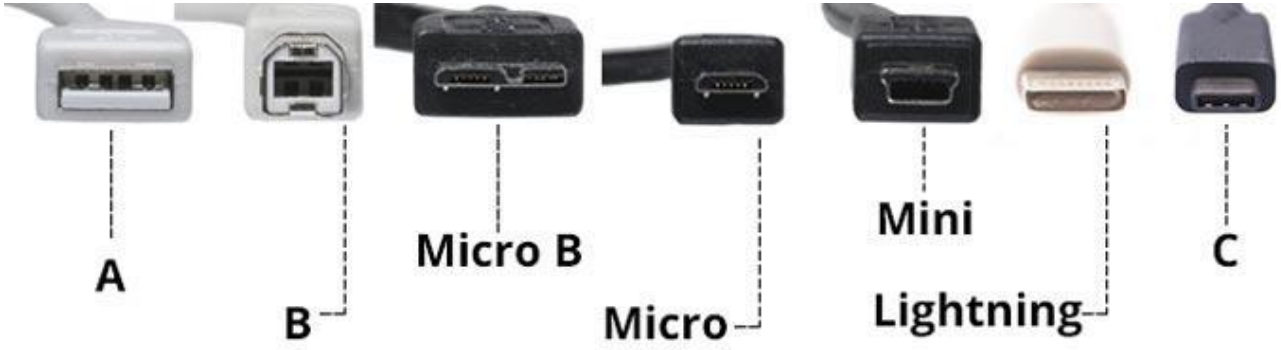


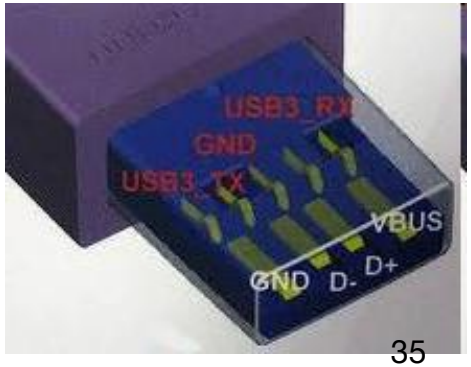


Interface Unit

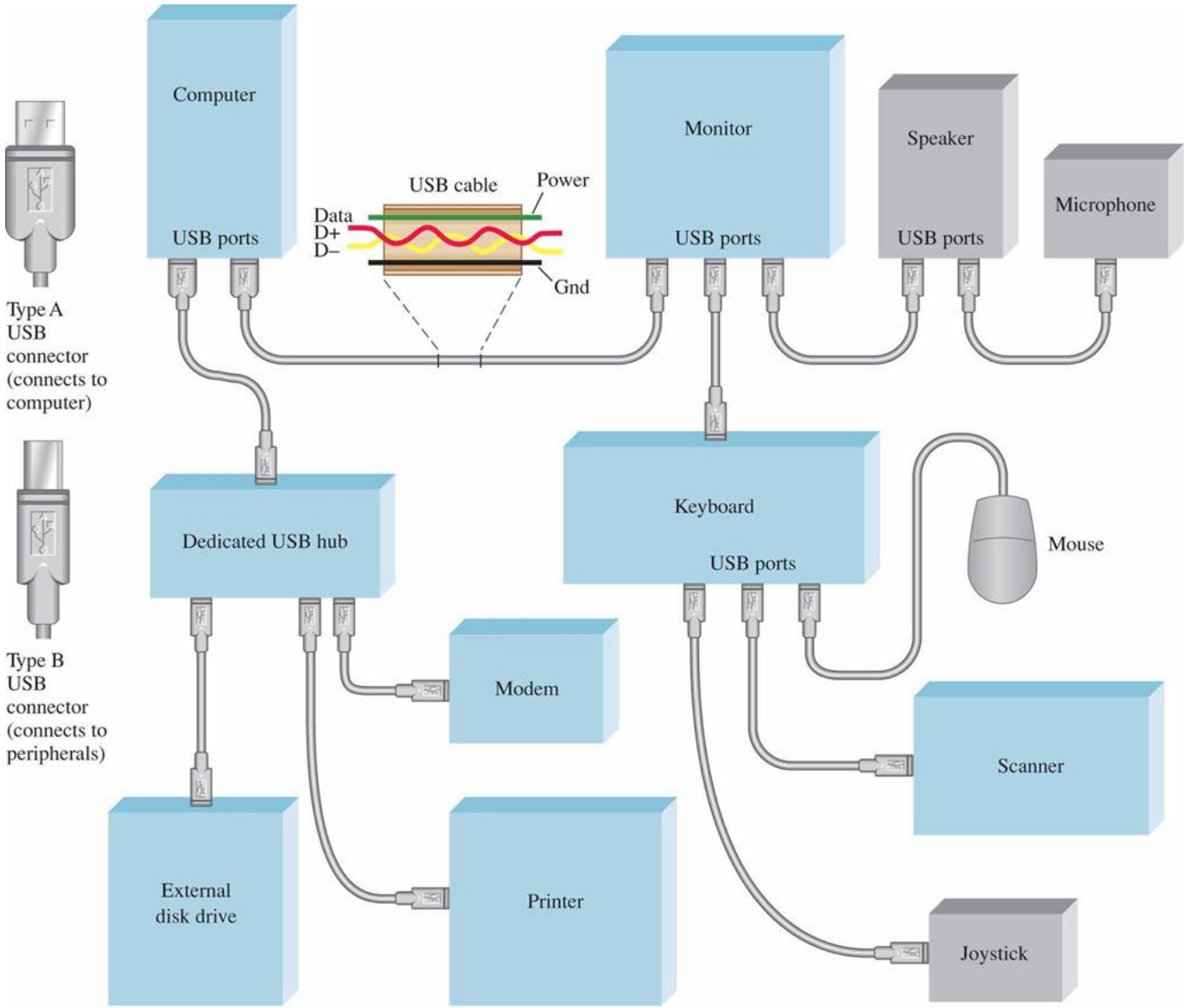
- EIA-232 (230kbps) / Centronics (1Mbps)
- USB 2.0 (Hi-Speed 480Mbps)
- USB 3.1(10Gbps)
- FireWire or i-Link (IEEE 1394) 400-3200Mbps
- Intel Thunderbolt (10Gbps)







Computer system with USB interfacing



Interface Unit

- PATA or IDE (Parallel Advanced Technology Attachment)
  - UDMA6 (133 MBs)



- SATA I (1.5), II (3.0), III (6.0Gbps)



- eSATA (2.4Gbps)
- SAS (Serial Attached SCSI 4.8Gbps) / Parallel SCSI (2.5Gbps)
- Fiber Channel (6.8Gbps)
- HDMI (High-Definition Multimedia Interface)
  - HDBaseT
  - HDMI 1.4 (10Gbps)



Secondary Memory / Storage Memory

- Magnetic Tape
- Floppy Disk
- Hard Disk
- Optical Disk
- Thumb Drive

The IBM Model 350 disk file with a storage space of 5MB from 1956 and a Micro SD Card

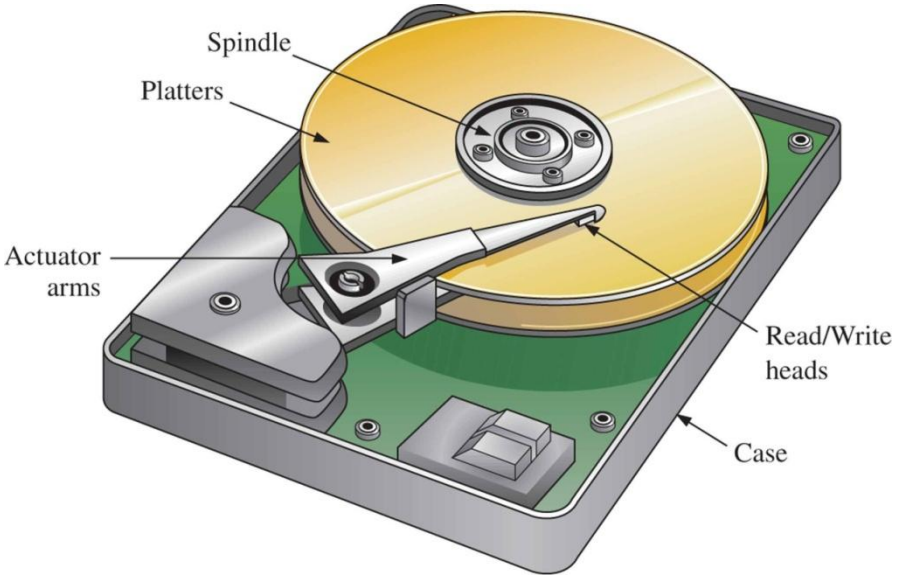


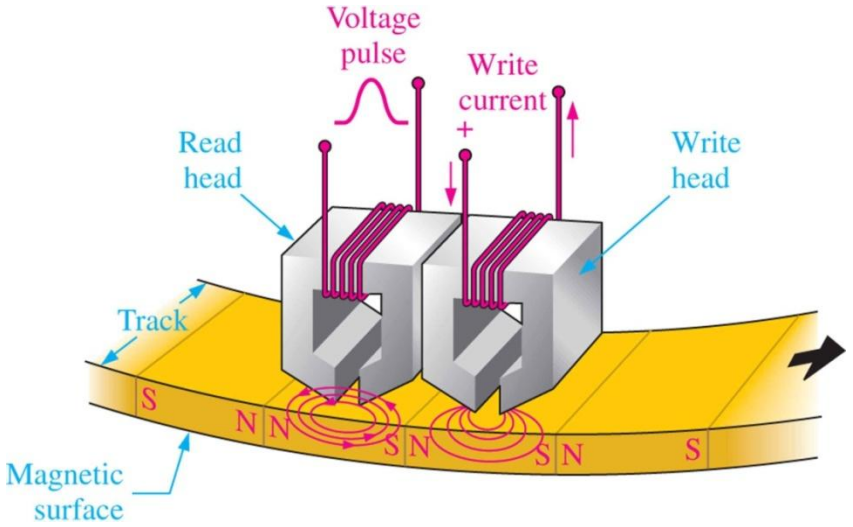


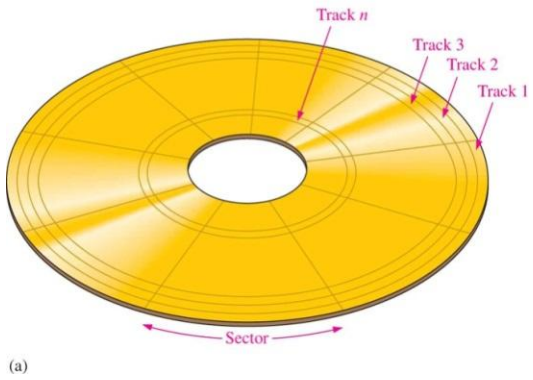


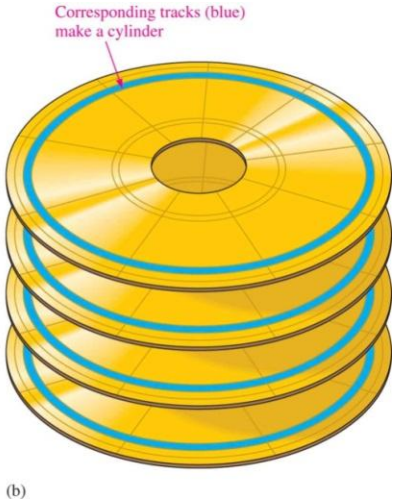


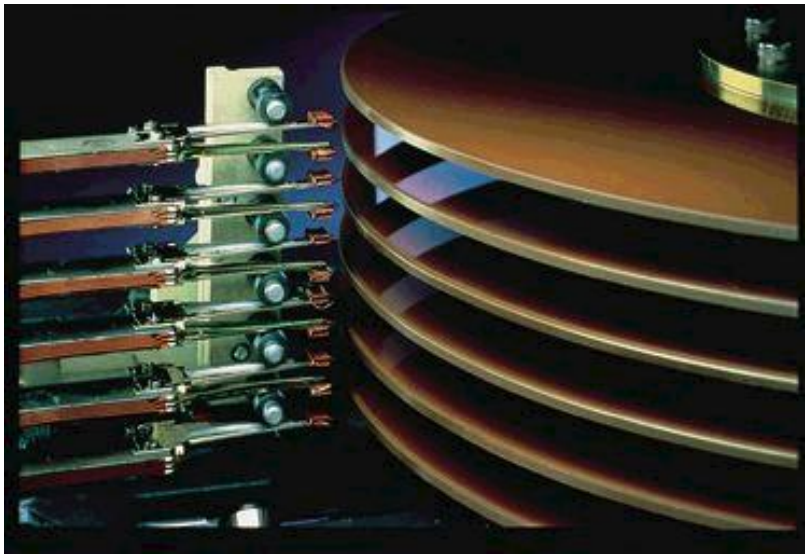
Hard Disk Drive (14TB)







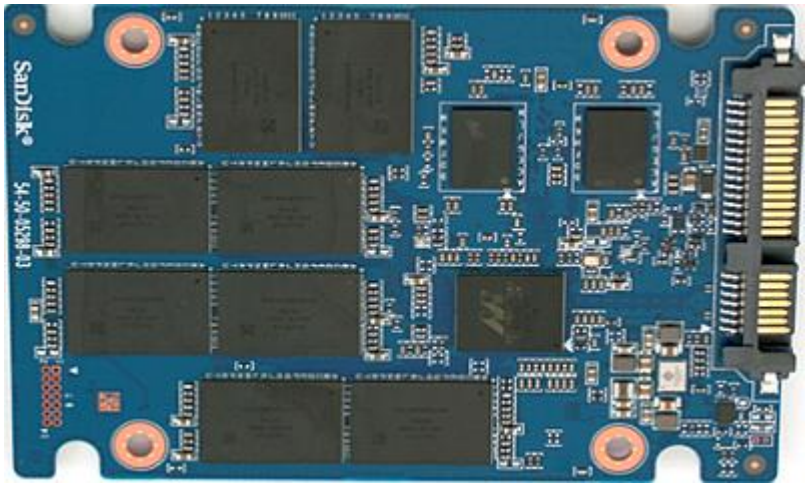




Solid-State Drive

- WD 2TB

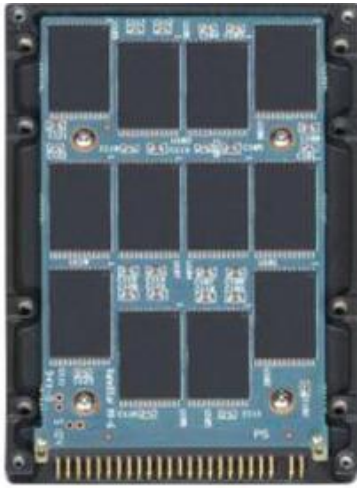








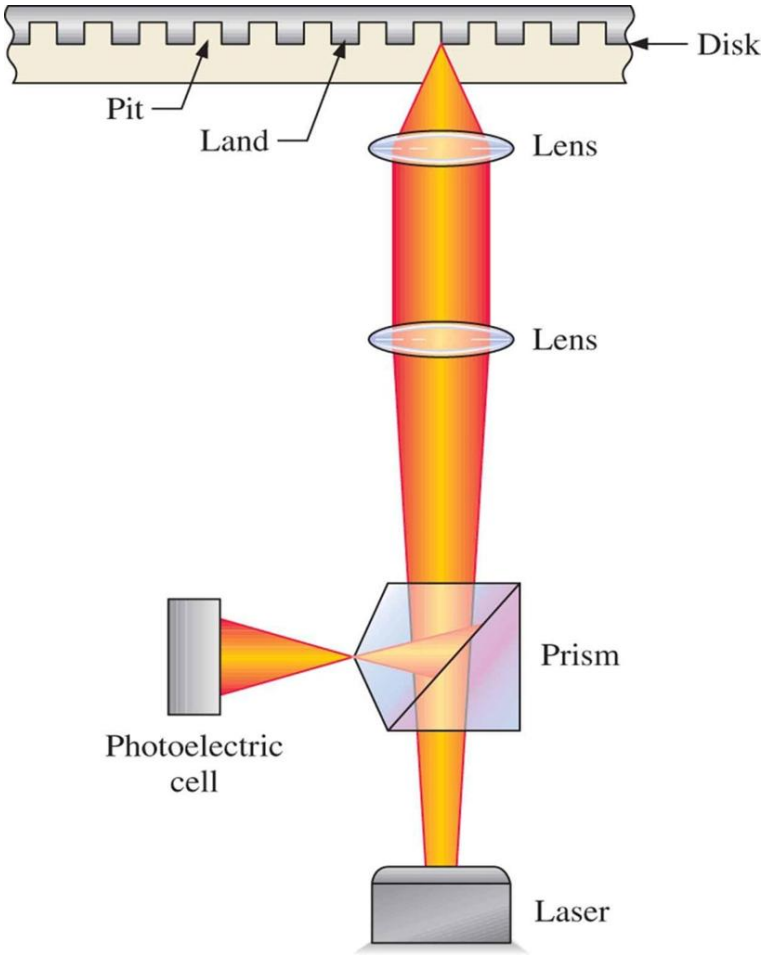
Traditional hard disk drive

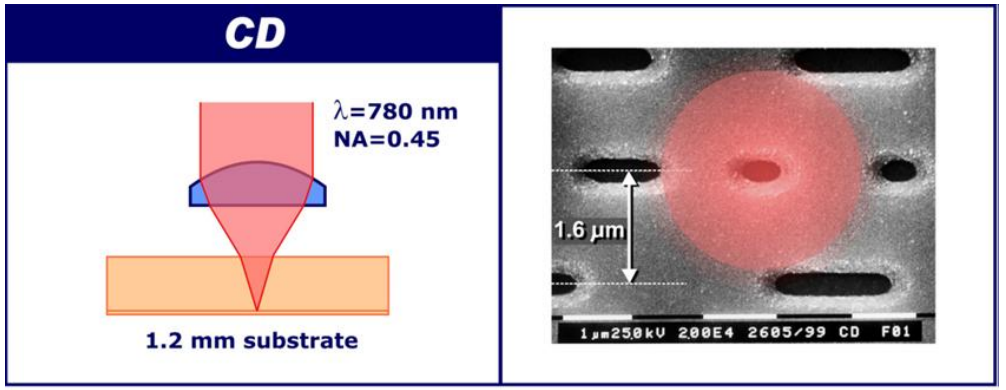


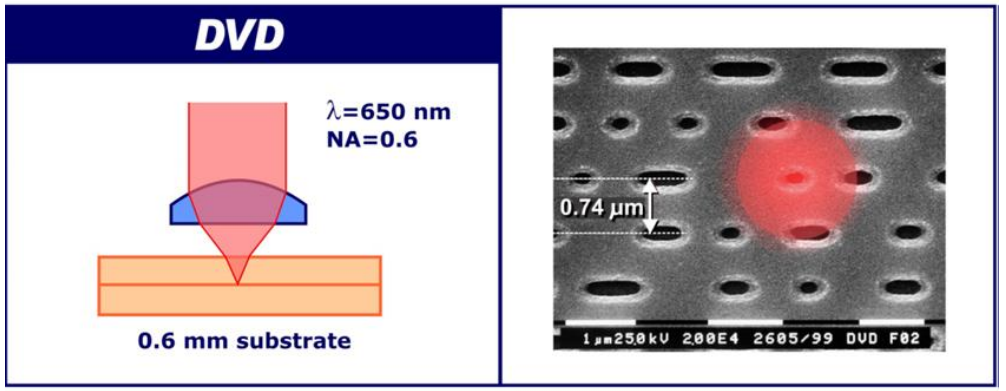
Solid state hard drive

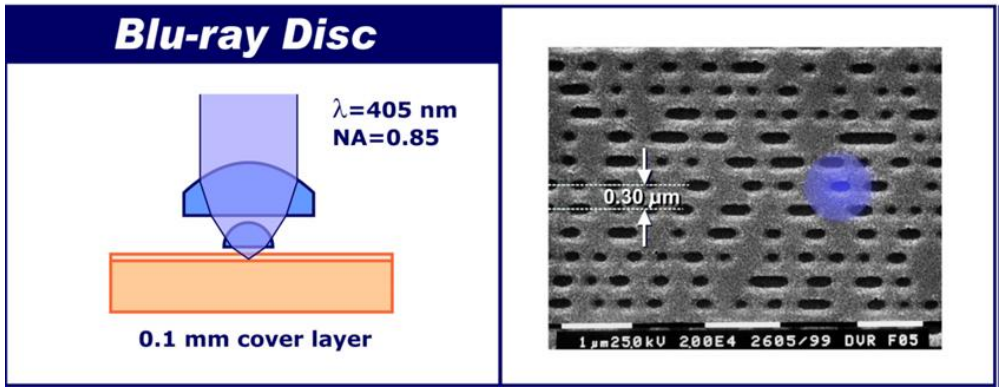
Optical Disc Drive - ODD

- CD-ROM / DVD
- Blu-Ray Disc (25/50GB)
  - BDXL (100/128 GB)

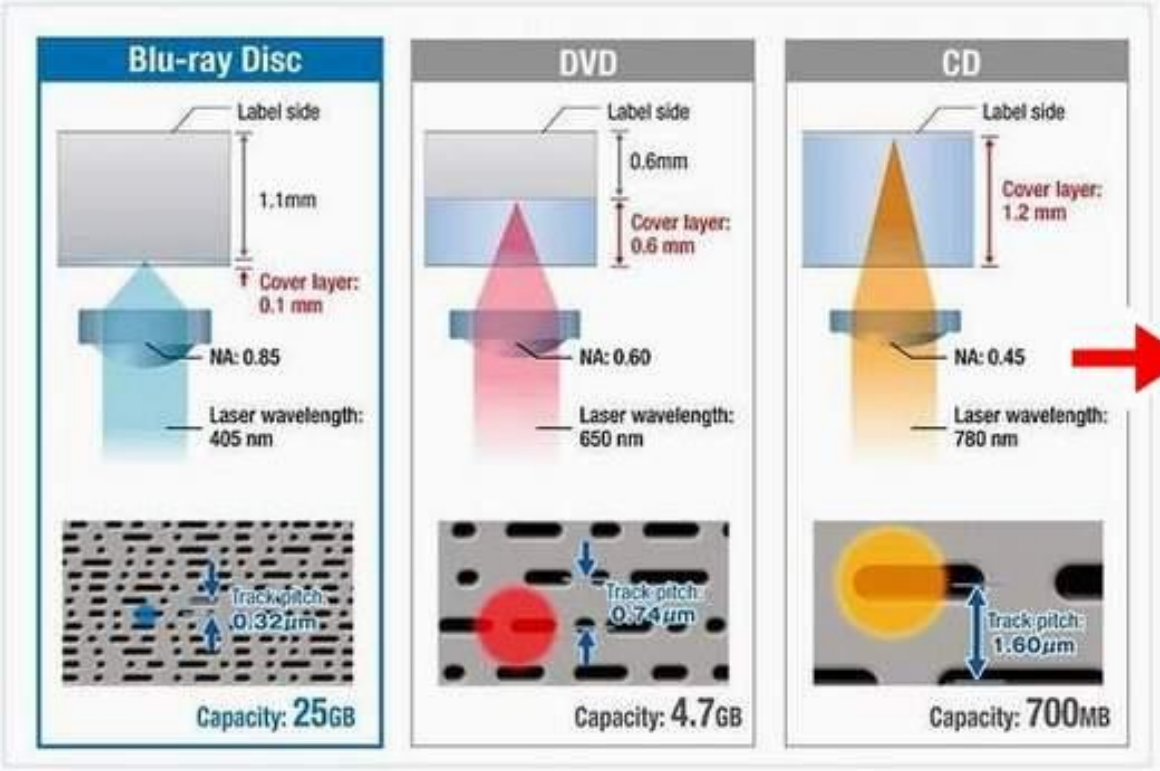








Storage Trend





Blu-ray Disc™

Feature Phone vs Smart Phone









HP Tablet, Nexus, iPad, DeX









